

Section I: English Part I

Questions on “Black Gold.” Write brief answers no more than 2 sentences

1. For what was oil first used?

Oil was initially used in burning lamps.

2. What is the purpose of the mud during the drilling operation?

It is used to control the well, lubricate the drill bit, and carry away rock particles.

3. What usually happens to the oil once it is brought to the surface?

It is pumped through pipelines to a refinery.

4. Name three of the environments where we look for oil.

The jungle, the desert, and the sea.

5. What is “Black Gold” and why is it given this name?

Black gold is oil. It is given this name because it is a precious commodity. It is a valuable natural resource with limited supply.

Section I: English Part II

1. B	6. D	11. D	16. B	21. B
2. C	7. C	12. B	17. C	22. C
3. B	8. B	13. A	18. A	23. D
4. B	9. C	14. D	19. B	24. C
5. B	10. A	15. C	20. B	25. D

Family Name _____

First Name _____

ANSWER SHEET PAGE 2 of 2

Answer Sheet for Mathematics, Mechanical, Electrical and General sections.
Mark all answers on this sheet.

Mathematics

1. B	6. B	11. C	16. B
2. A	7. D	12. B	17. D
3. A	8. B	13. C	18. B
4. C	9. C	14. A	19. B
5. C	10. A	15. D	20. C

Mechanical

1. C	4. B	7. B	10. C	13. D
2. A	5. B	8. B	11. A	14. D
3. C	6. C	9. B	12. C	15. D

Electrical

1. A	4. C	7. B	10. C
2. C	5. C	8. C	11. A
3. B	6. A	9. B	12. C

General

1. C	4. D	7. C	10. B	13. B
2. D	5. A	8. C	11. A	
3. C	6. D	9. B	12. C	

14. Safety Precaution Number 1
Fasten seat belts. (or something similar)

Safety Precaution Number 2

Lock doors. (or something similar)

Field Specialist Aptitude Test

Section I : English
Section II : Mathematics
Section III : Mechanical
Section IV : Electrical
Section V : General

Name :

Date :

**Technical :
College**

**Graduation :
Date**

***Please write all your answers on the answer sheets provided.**

Family Name _____

First Name _____

ANSWER SHEET PAGE 1 of 2**Section I: English Part I**

Questions on “Black Gold.” Write brief answers no more than 2 sentences

6. For what was oil first used?
7. What is the purpose of the mud during the drilling operation?
8. What usually happens to the oil once it is brought to the surface?
9. Name three of the environments where we look for oil.
10. What is “Black Gold” and why is it given this name?

Section I: English Part II

1.	6.	11.	16.	21.
2.	7.	12.	17.	22.
3.	8.	13.	18.	23.
4.	9.	14.	19.	24.
5.	10.	15.	20.	25.

Family Name _____

First Name _____

ANSWER SHEET PAGE 2 of 2

Answer Sheet for Mathematics, Mechanical, Electrical and General sections.
Mark all answers on this sheet.

Mathematics

1.	6.	11.	16.
2.	7.	12.	17.
3.	8.	13.	18.
4.	9.	14.	19.
5.	10.	15.	20.

Mechanical

1.	4.	7.	10.	13.
2.	5.	8.	11.	14.
3.	6.	9.	12.	15.

Electrical

1.	4.	7.	10.
2.	5.	8.	11.
3.	6.	9.	12.

General

1.	4.	7.	10.	13.
2.	5.	8.	11.	
3.	6.	9.	12.	

14. Safety Precaution Number 1

Safety Precaution Number 2

Section I: English Part I

Read the text below and then answer the questions that follow:

What is Black Gold?

Oil is such a precious commodity that it has been called “Black gold”. We depend upon it for products ranging from fuel to fertilizers. Demand is currently so high and increasing so rapidly that some experts predict oil supplies will dry up within forty years. If that happens man will have managed in a little more than one hundred and fifty years to use up a gift that took nature several million years to create.

Oil was initially used in burning lamps to provide light for homes, offices and factories. Only after the invention of the internal combustion engine, as used in cars, did the modern Oil Industry develop as we know it.

The never ending search for “Black Gold” takes people and equipment all over the world. The only real way of knowing whether oil is likely to be present in the ground is to drill a hole which is often called an exploration or test well. This can sometimes mean drilling more than 6000 meters beneath the earth’s surface. The cost of such an exercise is high and it can run into millions of dollars and sometimes the results can be disappointing. A well in Texas which went to 7,600 meters cost nearly 3.0 million dollars and did not produce a drop of oil.

Mud is an all important component in the drilling process. It is used to control the well, lubricate the drill bit and carries away rock particles which can be gathered at the surface and examined.

Once oil is found then it is usually pumped through pipelines to a refinery where it is separated into various products.

Anybody working for an international oil related company may find themselves in a country where language, climate, food, customs and habits are totally different to their own. Work locations may be extreme; the jungle, the desert and the sea are often explored. Such are the many challenges for a man and his machine in the search for Black Gold.

Section I: English Part II

For questions 1 to 10, select the correct answer:

1. He is not only reliable, but also _____.
 - A. is intelligent and creating
 - B. intelligent and creative
 - C. intelligence and creating
 - D. to be intelligent and creative

2. If you _____, we wouldn't have missed the train.
- A. didn't come late
 - B. wouldn't have been late
 - C. hadn't been late
 - D. were not late
3. The manager _____ here at 4 p.m.
- A. is supposing to arrive
 - B. is supposed to arrive
 - C. will suppose to arrive
 - D. is supposed to be arrived
4. Due to the devaluation, many people have become rich, but many others _____.
- A. ruined
 - B. have been ruined
 - C. have ruined
 - D. would have ruined
5. The man has been hospitalized since yesterday _____.
- A. because his serious wound
 - B. because of his serious wound
 - C. because of his seriously wound
 - D. because his wound seriously
6. Your suggestion, _____, is beneficial to both the company and the employees.
- A. Different than his
 - B. as well as he
 - C. as he is
 - D. like his
7. Many people _____ for six hours without stopping.
- A. can't stand for doing a test
 - B. can't stand to be doing a test
 - C. can't stand doing a test
 - D. can't stand a test to be done
8. I don't know _____.
- A. where is your pen
 - B. where your pen is
 - C. where your is pen
 - D. where is the pen of yours

9. If you want to see him, _____.

- A. please let me to know
- B. let me to know please
- C. please let me know
- D. let me please to be known

10. _____, the little girl ran out of the yard.

- A. Being afraid of the dog
- B. The dog she is afraid of
- C. The dog has been afraid of
- D. Was being afraid of the dog

For questions 11 to 15, select the correct preposition or adverb:

11. He insisted _____ seeing the documents.

- A. by
- B. in
- C. of
- D. on

12. It never occurred _____ me to ask him for proof of his identity.

- A. for
- B. to
- C. on
- D. with

13. I object to being kept waiting. Why can't you be _____ time?

- A. on
- B. in
- C. of
- D. at

14. I was _____ the impression that I had paid you for the work you did for me.

- A. into
- B. over
- C. with
- D. under

15. Buying presents for children is sometimes very difficult. _____ the end I bought a kite for Tom.

- A. At
- B. On
- C. In
- D. By

For questions 16-20, choose the correct tense for the verbs in brackets:

16. I usually go by train, but this weekend I am going by bus. It (take) longer but it costs less.

- A. take
- B. takes
- C. is taken
- D. will take

17. I (not think) your brother is enjoying the party. He keeps looking at his watch.

- A. doesn't think
- B. didn't think
- C. don't think
- D. am not thinking

18. The hall (be) painted at the moment, so it isn't looking its best.

- A. is being
- B. is
- C. is to be
- D. will be

19. At present I'm having my typewriter (repair).

- A. repairing
- B. repaired
- C. is repaired
- D. being repaired

20. I (interest) in learning foreign languages.

- A. interested
- B. am interested
- C. am interesting
- D. interesting

For questions 21 to 25, four words or phrases have been underlined in each sentence. Please choose the one word or phrase, which is **incorrect** in standard written English.

21. To John, as well as I, it is extremely beneficial to make friends with people from
A B C
other countries and to ask them about their cultures.
D

22. Before making a plan for the weekend, most people listen to the weather forecast
A B
even though they know hardly nothing about the factors that affect the weather.
C D

23. The circumstances under which the operation took place were unbelievable in that
A B C
the doctor didn't even using antiseptic.
D

24. She was probably so disappointed in my not keeping the appointment and she
A B C
didn't even greet me when I came the following day.
D

25. Had he followed the doctor's advice and taken all the medicine regularly, the
A B C
doctor would not have looked angry at him.
D

Section II: Mathematics (Make sure to answer using the correct UNITS)

Conversion Units:

π	=	3.1416
1ft	=	30.48 centimeters
1 ft	=	12 inches
1kg	=	2.2 lbs
1 in	=	2.54 centimeters
1 gal	=	3.785 liters
1 ft ³	=	7.48 gal
1 bbl	=	42 gal
1 bbl	=	5.614 ft ³
1 psi	=	.145 kPa
weight of fresh water		8.33 ppg
Area of a circle	=	πr^2
Volume of cylinder	=	$\pi r^2 h$

- What is the area of a rectangle with dimensions of 6ft by 4ft?
A. 10 ft² B. 24 ft² C. 2 ft² D. 1.5 ft²
- What is the area of a right-angle triangle with dimensions of 3ft by 3ft?
A. 4.5 ft² B. 9 ft² C. 1 ft² D. 18 ft²
- Given the diameter of a circle is 10 centimeters, what is the surface area in cm²?
A. 78.5 cm² B. 2.5 cm² C. 62.8 cm² D. 100 cm²
- What is the total capacity in gallons of a rectangular tank 5 meters in height knowing that it will hold 200 liters per foot?
A. 3281 gal B. 1000 gal C. 867 gal D. 8670 gal
- What is the volume in cubic meters of a cylinder with a height of 2 meters and a diameter of 1.5 meters?
A. 14.1 m³ B. 4.5 m³ C. 3.5 m³ D. 18.8 m³
- How much time in minutes will it take to fill a tank that can hold 660 gallons if we are filling it at 33 gallons per hour?
A. 21780 minutes B. 1200 minutes C. 20 minutes D. 0.3 minutes

7. What is the force in pounds exerted upwards on a piston with a diameter of 4 inches if a gauge indicates the internal pressure in the cylinder is 1000 psi?
- A. 4000 lbs B. 250 lbs C. 50,266 lbs D. 12,566 lbs
8. What is the weight in kilograms of a string of pipe that is 11,500 ft long and weighs .33 pounds per inch?
- A. 1,725 kg B. 20,700 kg C. 45,540 kg D. 3795 kg
9. How much time in minutes will it take a car driving at 90 km/hr to travel 27 kilometers?
- A. 3.3 minutes B. 2430 minutes C. 18 minutes D. 0.3 minutes

Questions 10 - 15 refer to the following:

A tank is a vertical cylinder with inside diameter of 6ft and a height of 15ft:

10. What is the volume of liquid in cubic feet if we fill the tank to one foot high?
- A. 28.27 ft³ B. 18.8 ft³ C. 113.1 ft³ D. 424.1 ft³
11. What is the total volume of the tank in cubic feet?
- A. 1686.5 ft³ B. 442.5 ft³ C. 424.1 ft³ D. 847.5 ft³
12. What is the volume of water in cubic ft in the tank if a measure stick indicates the water surface is 6 feet below the top of the tank?
- A. 170.2 ft³ B. 254.4 ft³ C. 1017.9 ft³ D. 678.6 ft³
13. What is the weight in pounds of the tank, as in the above case, if the tank itself weighs 20 lbs per ft of height?
- A. 63,424 lbs B. 42,241 lbs C. 16,151 lbs D. 10,605 lbs
14. How much water in gallons would have to be removed to reduce the total weight of the tank plus water, as in the above case, by 20%?
- A. 388 gals B. 3,230 gals C. 2,121 gals D. 626 gals

15. What would be the new level of the water, as measured from the top of the tank, if the water in the above case (equivalent of 20% of total weight) is removed?

- A. 10.4 ft B. 12.7 ft C. 7.1 ft D. 7.8 ft

16. If $x + 14 = 32$, then $x = ?$

- A. 46 B. 18 C. 2.3 D. 0

17. If $(2x^2 - 2) 2 = 12$, then $x = ?$

- A. -2 B. +2 C. 4 D. A and B

18. If $x = y - 9$, and $y - 4 = 10$, $x = ?$

- A. -5 B. +5 C. 1 D. A and B

19. Perform the following operation:

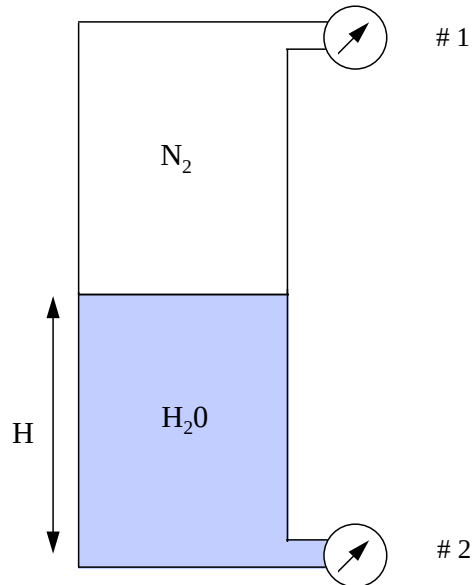
$$\frac{(a+b)}{(a-b)} \times \frac{(a+b)}{(a+b)} =$$

- A. $\frac{a^2 + b^2}{a^2 - b^2}$ B. $\frac{a^2 + 2ab + b^2}{a^2 - b^2}$ C. $\frac{a^2 + 2ab + b^2}{a^2 + b^2}$ D. $\frac{a^2 - 2ab - b^2}{a^2 + b^2}$

20. Find the equation of a straight line passing through the point A (3,2) and having a gradient of $m=5$.

- A. $y = 3x - 2$ B. $y = 5x + 5$ C. $y = 5x - 13$ D. $y = 2x + 3$

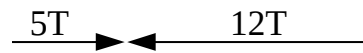
Section III: Mechanical



1. For the tank shown above what is the pressure in psi measured at gauge #2 if: the pressure at gauge #1 is 300 psi (21 kg/cm^2) and the level of the water (H) is 70 ft (21.4 m).

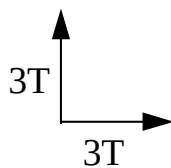
A. 315 psi B. 23 kg/cm^2 C. 330 psi D. 300 psi

2. Indicate the resultant force, magnitude, and direction:

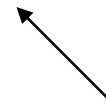


A. $\leftarrow 7T$ B. $\rightarrow 7T$ C. $17T \rightarrow$ D. $\leftarrow 17T$

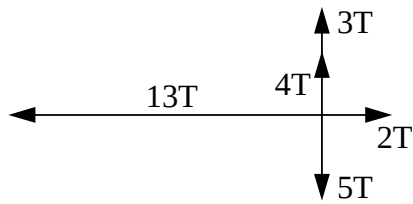
3. Indicate the resultant force, magnitude and direction:



A. 4.24 T B. 9 T C. 4.24 T D. 9 T

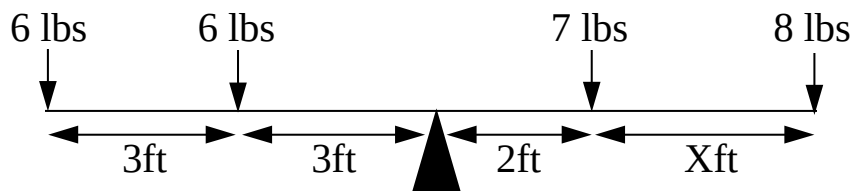


4. Indicate the resultant force, magnitude and direction:



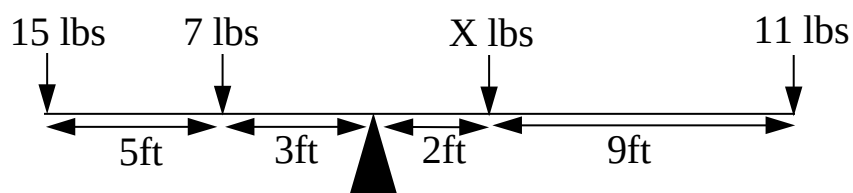
- A. 27 T B. 11.18 T C. 13.28 T D. 6.22 T
- ← ↙ ↗ →

5. For the system shown below, what should the distance X be for the system to be in equilibrium?



- A. 2.75 ft B. 3.0 ft C. 3.25 ft D. 2.5 ft

6. For the system shown below, what force should be applied at X for the system to be in equilibrium?

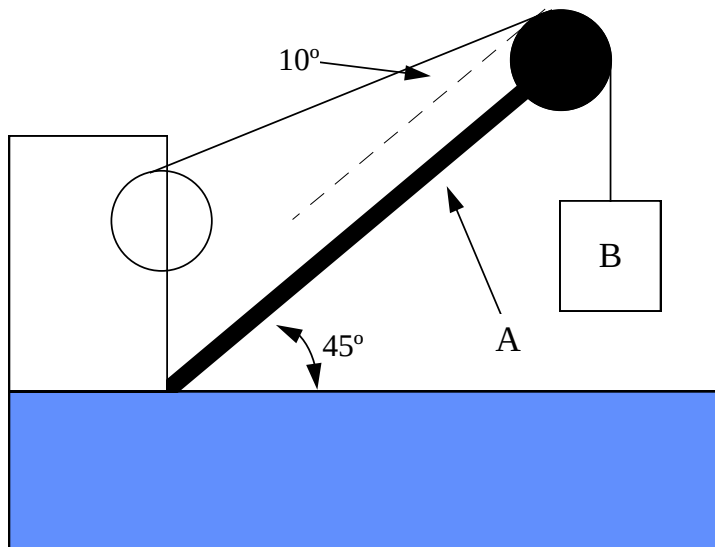


- A. 8 lbs B. 9 lbs C. 10 lbs D. 11 lbs

7. For any given lever, the shorter the lever, the greater the lifting power.

- A. True B. False

8. A body is floating in seawater with 2/3 of its volume above the surface. The weight of the displaced water is?
- A. Less than the weight of the body
 - B. Equal to the weight of the body
 - C. More than the weight of the body
 - D. Not enough information provided
9. Consider the crane lifting the load shown. If the maximum crane loading on member A is 40,000 lbs (18,140 kg) as a bending limit, calculate the heaviest load that can be lifted.



$$\cos 45 = .707$$

$$\sin 45 = .707$$

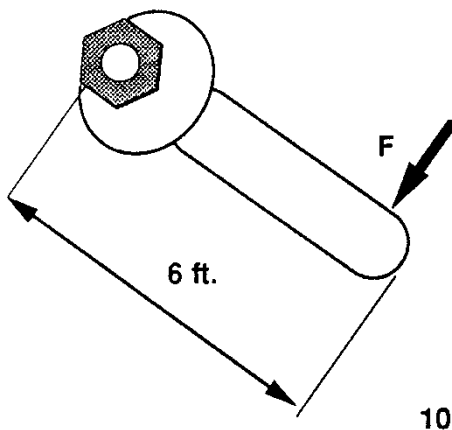
$$\cos 10 = .985$$

$$\sin 10 = .174$$

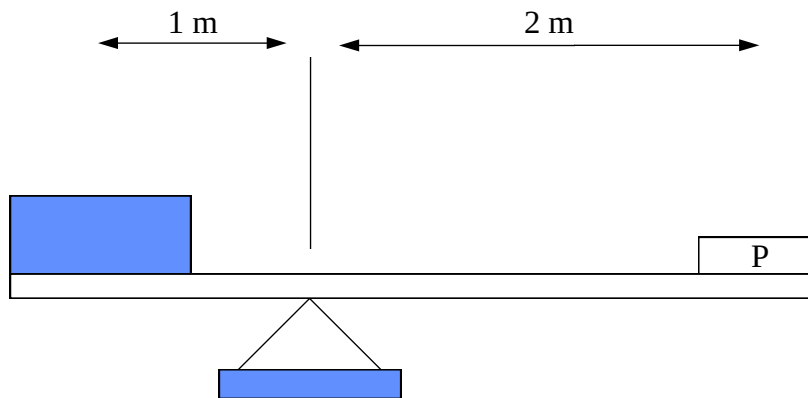
- A. 34,112 lbs
- B. 75,046 lbs
- C. 49,140 lbs
- D. 28,280 lbs

10. What is the value of F if the torque is 50 lbs/ft ?

- A. 100 lbs
- B. 300 lbs
- C. 8.33 lbs
- D. 25 lbs



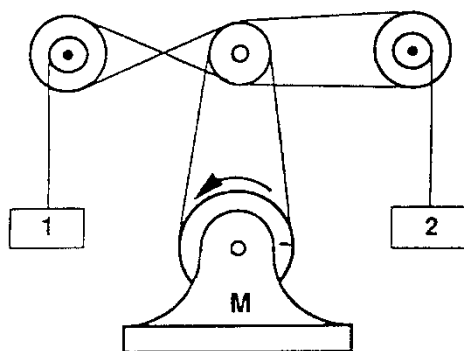
11. For the system to be in equilibrium, what is the mass of P in pounds? The mass at the other end of the beam is 5 kg.



- A. 5.5 lb B. 2.8 kg C. 10 kg D. none of these

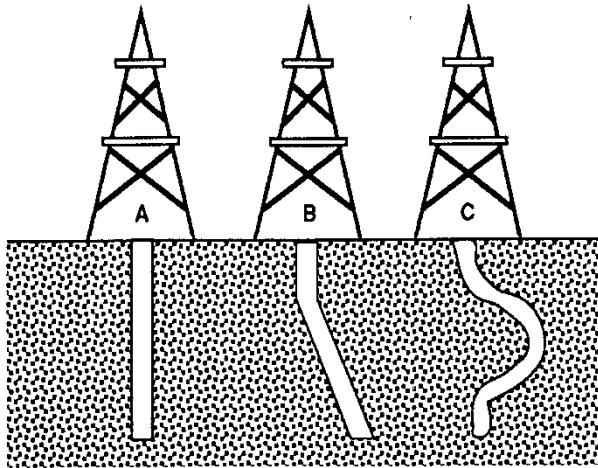
12. If the motor M rotates in the direction shown by the arrow, what is going on ?

- A. 1 and 2 are going down
 B. 1 goes up and 2 goes down
 C. 1 and 2 are going up
 D. 1 goes down and 2 goes up



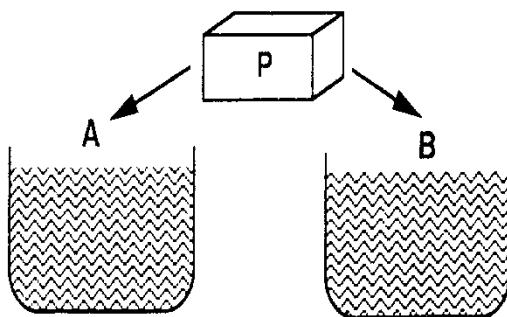
13. These 3 wells have been drilled with a mud of specific gravity of 1.25 gr/cm^3 . In which well A, B or C, will the highest hydrostatic bottom hole pressure be recorded ?

- a. Well A
- b. Well B
- c. Well C
- D. All three will have equal bottom hole pressure



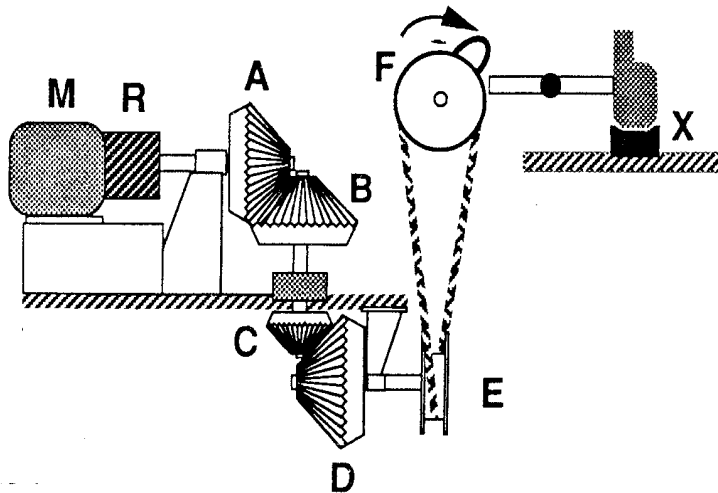
14. If the density of A is greater than the density of B. What is going to happen ?

- A. If P floats in liquid A, it will also float in B.
- B. If P floats in liquid B, it may or may not float in A, depending on the density of P.
- C. If P sinks in B, it will sink in A also.
- D. None of the above answers are correct.



15. The bevel gear A has 70 teeth, bevel gear B 140, C 114, and D 456. The diameter of the pulley E = $\frac{1}{2}$ diameter of pulley F. At what speed in RPM must rotate the motor M to allow the hammer to knock anvil X twice per second, taking into consideration the gear box R whose ratio inlet/ outlet is 5/1?

- A. 2400
- B. 1920
- C. 7200
- D. 9600



Section IV: Electrical

1. What is the equation for Ohm's law?

- A. $V = IR$ B. $V = I/R$ C. $R = VI$ D. $I = VR$

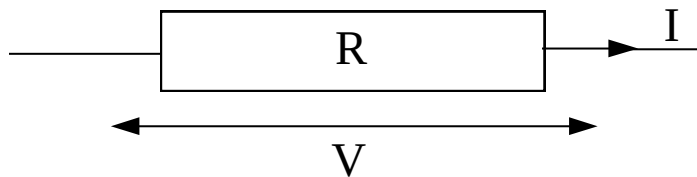
2. In direct current $I = 3A$ $R = 4\Omega$ Calculate V

- A. 1.33 V B. 6 V C. 12 V D. 24 V

3. What becomes V if we use 2 resistors of 4Ω in parallel?

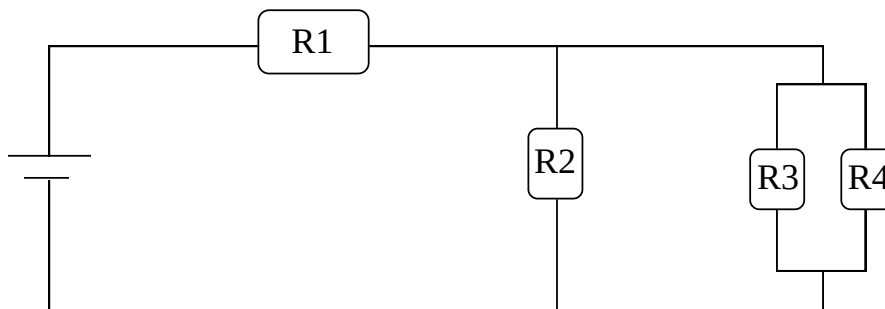
- A. 2.66 V B. 6 V C. 12 V D. 24 V

4. If the resistor below is 100 ohms and the voltage drop is 220 volts, what is the current I ?



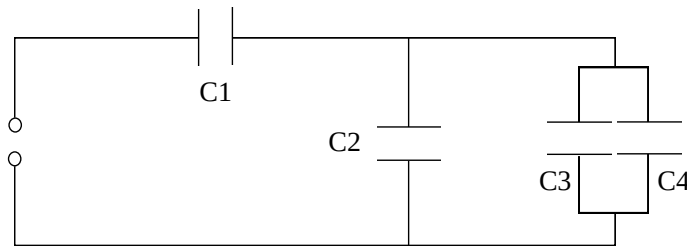
- A. .22 A B. 22 A C. 2.2 A D. 2.0 A

5. If $R_1=2k$ ohms, $R_2=1k$ ohms, $R_3=2k$ ohms and $R_4=2k$ ohms, what is the equivalent resistance of the circuit?

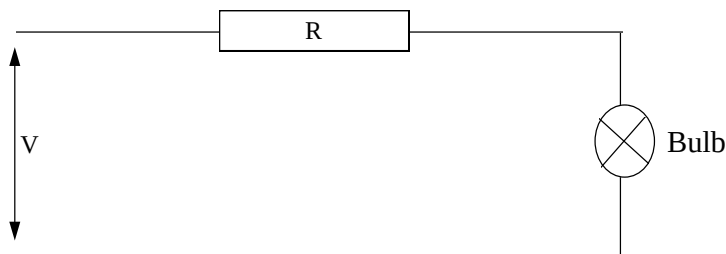


- A. 2250 ohms B. 1500 ohms C. 2500 ohms D. 2000 ohms

6. If $C_1=6F$, $C_2=1F$, $C_3=1F$ and $C_4=1F$, what is the equivalent capacitance of the circuit shown below?



- A. 2F B. 1.75 F C. 2.25 F D. 2.125 F
7. To reverse the direction of rotation in a 3hp, AC motor we must:
- A. Change round all three wires cyclically.
 B. Change round any two wires
 C. Cannot change direction of rotation.
 D. Change switch position, ie “on” de-energized and “off” energized to get the reversed direction.
8. For the circuit shown below, if $V=380VAC$ and the bulb is rated at $170VAC$, $5W$, what is the resistance R that should be connected in series to the bulb?

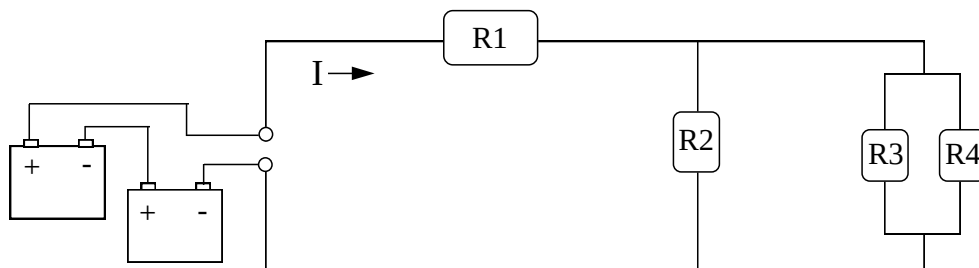


- A. 6500 ohms B. 7000 ohms C. 7145 ohms D. 5780 ohms
9. If we increase the voltage across a DC motor, then the motor speed will?
- A. decrease B. increase C. stay same D. stop
10. What will be the output voltage of two 12V batteries connected in series?
- A. 12 V B. 6 V C. 24 V D. no voltage

11. What will be the output voltage of two 12V batteries connected in parallel?

- A. 12 V B. 6 V C. 24 V D. no voltage

12. In the circuit shown below what is the current if the batteries are 12V each, $R_1=20$ ohms, $R_2=10$ ohms, $R_3=20$ ohms, and $R_4=20$ ohms?

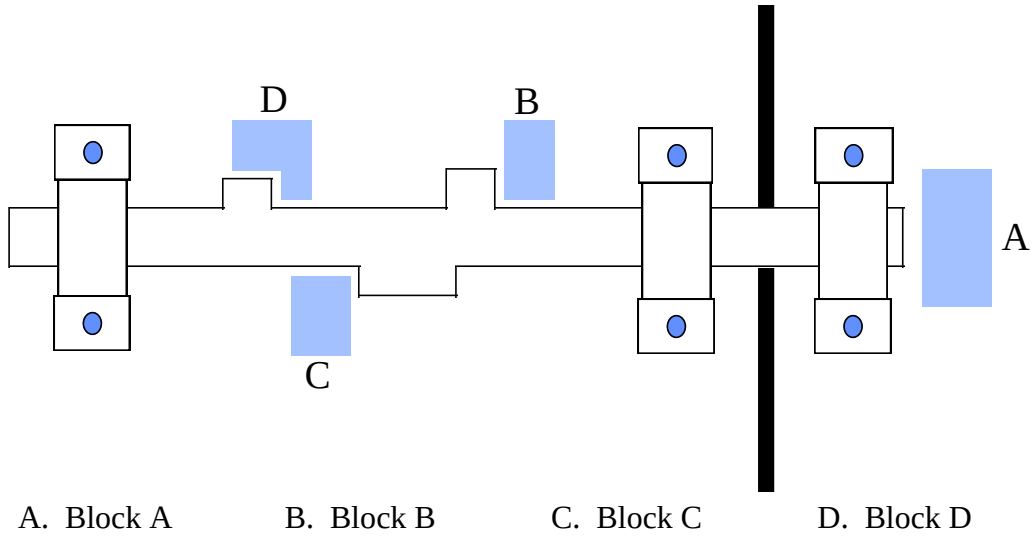


- A. 1000 mA B. 660 mA C. 960 mA D. 1200 mA

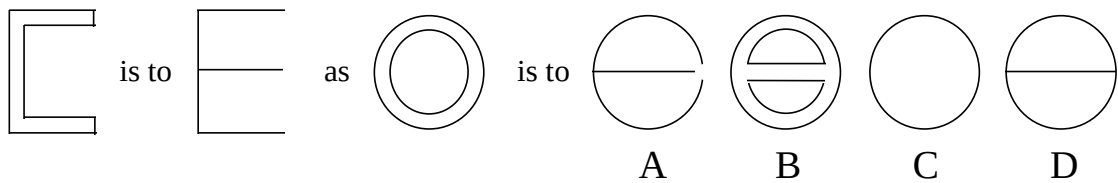
Section V: General

Find the right choice in the following:

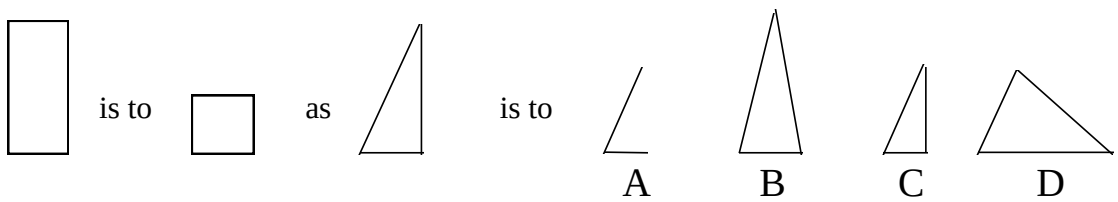
1. Which block has to be removed to be able to pull the bolt to the left and open the door?



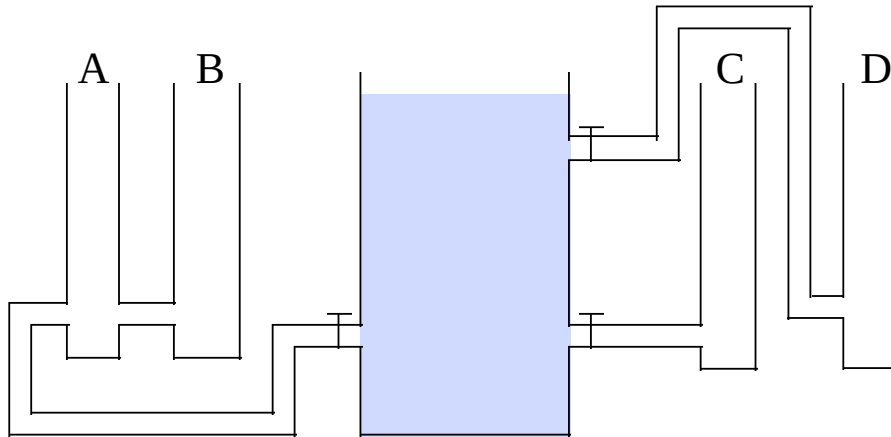
- 2.



- 3.

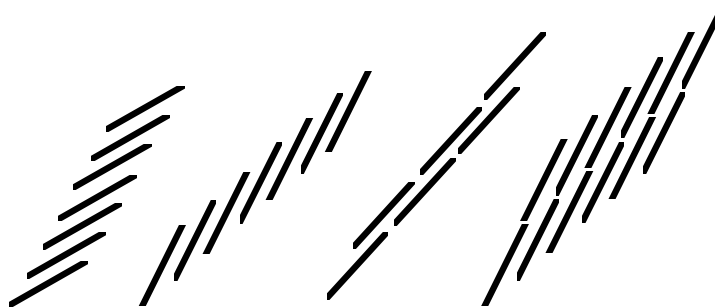


4. Into which container will the water **not** flow if we open all the taps?



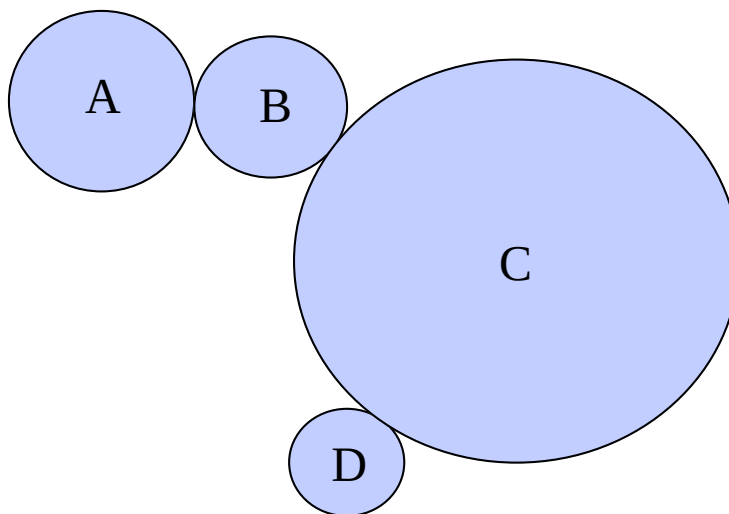
- A. Choice A B. Choice B C. Choice C D. Choice D

5. Which arrangement of tiles would stop rain from penetrating?



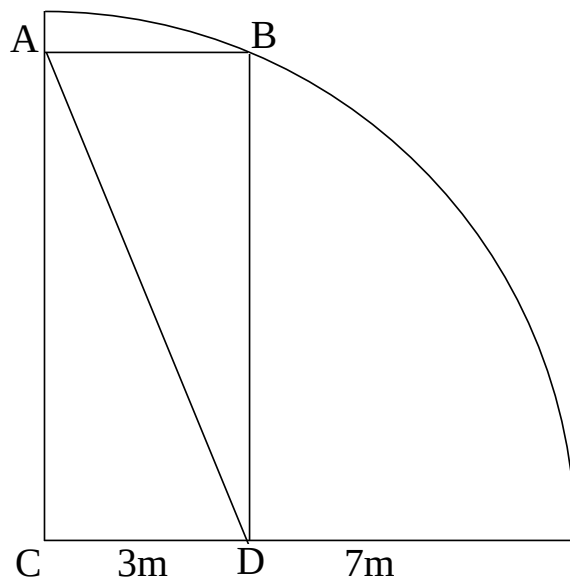
- A. Choice A B. Choice B C. Choice C D. Choice D

6. Which gear will make the most turns in one minute?

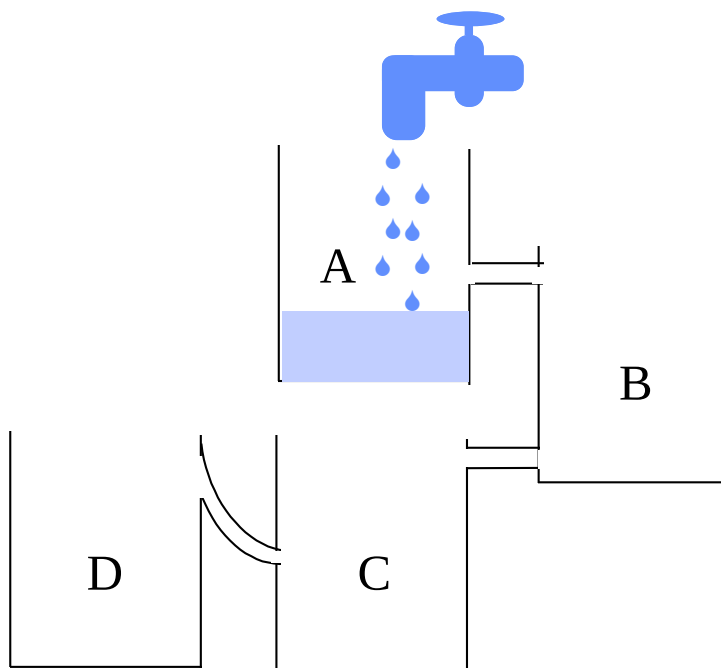


- A. Choice A B. Choice B C. Choice C D. Choice D

7. A quadrant contains an inscribed rectangle ABCD as shown. Given the distances marked, what is the length of AD?

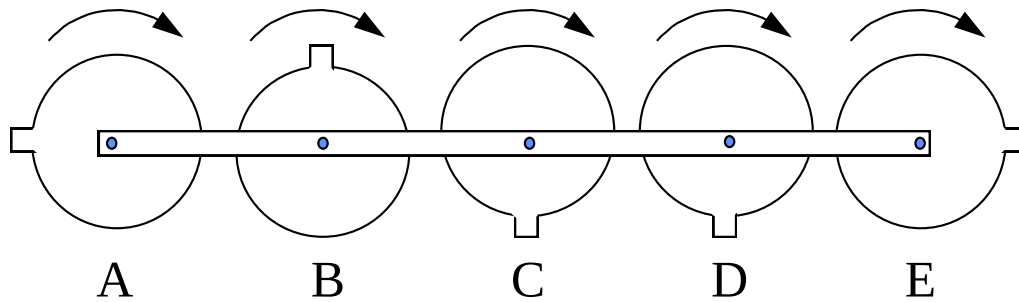


- A. 3 m B. 7 m C. 10 m D. 13 m
8. Which container will fill first?



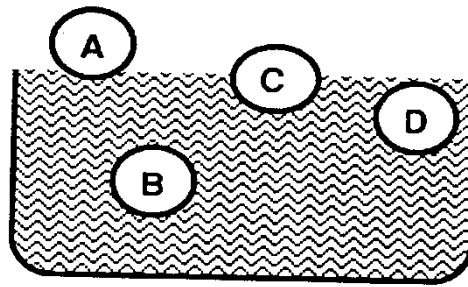
- A. Choice A B. Choice B C. Choice C D. Choice D

9. The five cogs are turning in the same direction and at the same speed. Which two will clash at a given moment?



- A. A and B B. B and C C. C and D D. D and E
10. In computer jargon, what do you understand by RAM?
- A. Read Access Memory
B. Random Access Memory
C. Real Access Memory
D. Random Acquired Memory
11. In computer jargon, what do you understand by ROM?
- A. Read Only Memory
B. Random Only Memory
C. Real Output Memory
D. Read Output Memory
12. In computer jargon, what do you understand by CPU?
- A. Computer Processing Unit
B. Computer Procedure Utility
C. Central Processing Unit
D. Central Power Unit

13. There are 4 balls of the same size in a container filled with water. Which ball is the most dense?



- A. Choice A B. Choice B C. Choice C D. Choice D

14. What are two safety precautions that should be taken before driving your car?
(Answer in your own words on the Answer Sheet)